Step 2 : Data

The given sample information are

City	Sample Size	Sample Proportion
City-1	$m = 500$	$p_X = \frac{400}{500} = 0.80$
City-2	$n = 800$	$p_Y = \frac{600}{800} = 0.75$

Here, $m \geq 30$, $n \geq 30$, $mp_X = 400 > 5$, $m(1-p_X) = 100 > 5$, $np_Y = 600 > 5$ and $n(1-p_Y) = 200 > 5$.

Step 3 : Level of significance

$\alpha = 5\%$

Step 4 : Test statistic

The test statistic under the null hypothesis is

$$Z = \frac{p_X - p_Y}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{m}+\frac{1}{n}\right)}} \text{ where } \hat{p} = \frac{mp_X + np_Y}{m+n} \text{ and } \hat{q} = 1-\hat{p}$$

The sampling distribution of Z under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculation of Test Statistic

The value of Z for given sample information is calculated from

$$z_0 = \frac{p_X - p_Y}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{m}+\frac{1}{n}\right)}}.$$

Now, $\hat{p} = \frac{400+600}{500+800} = \frac{1000}{1300} = 0.77$ and $\hat{q} = 0.23$

Thus, $z_0 = \frac{0.80-0.75}{\sqrt{(0.77)(0.23)\left(\frac{1}{500}+\frac{1}{800}\right)}}$

$z_0 = 2.0764$

Step 6 : Critical value

Since H_1 is a two-sided alternative hypothesis, the critical value at 5% level of significance is $z_e = 1.96$.

Step 7 : Decision

Since H_0 is a two-sided alternative, elements of the critical region are determined by the rejection rule $|z_0| > z_e$. Thus, it is a two-tailed test. For the given sample information, $z_e = 2.0764 > z_e = 1.96$. Hence, H_0 is rejected. We can conclude that the difference between the proportions of self-employed people in City-1 and City-2 is significant.

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